PI

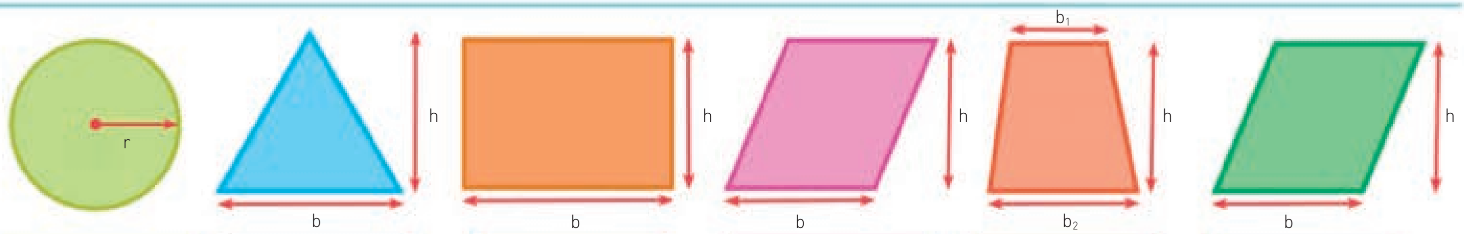
If you divide the circumference of a circle by its diameter, the answer is always 3 and a bit, or pi (π). It is impossible to write pi precisely, because the numbers after the decimal point continue forever.

$$\pi = 3.141592653$$

PI SYMBOL

FINDING AREA

The area of a two-dimensional shape is the amount of space inside it. There are formulae that can be used to work out how much space there is inside any polygon.



area = πr^2

CIRCLE
The area of a circle is pi (3.14) multiplied by the square of the circle's radius.

area = $\frac{1}{2}$bh

TRIANGLE
To find the area of a triangle, multiply the base by the height, then halve your answer.

area = bh

RECTANGLE
The area of a rectangle can be found by multiplying its base by its height.

area = bh

PARALLELOGRAM
Find the area of a parallelogram by multiplying its base by its vertical height.

area = $\frac{1}{2}h(b_1+b_2)$

TRAPEZIUM
Find the area by adding the two parallel sides, multiplying the total by the height, then dividing by 2.

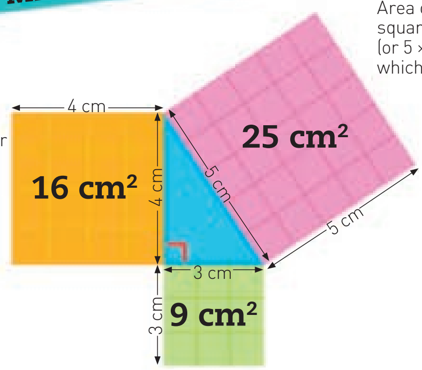
area = bh

RHOMBUS
The area of a rhombus can be found by multiplying its base by its vertical height.

PYTHAGORAS'S THEOREM

This theory is named after an Ancient Greek mathematician called Pythagoras. He observed that if you draw squares from each side of a right-angled triangle, the area of the two smaller squares added together is equal to the area of the largest square.

"GEOMETRY" COMES FROM GREEK: "GEO" MEANING EARTH AND "METRIA" MEANING MEASURE



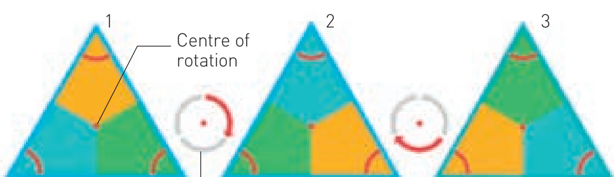
USING THE THEOREM
Pythagoras's theorem can be used to find the length of the longest side of a right-angled triangle (c), if you know the length of the two shorter sides (a and b).

$a^2 + b^2 = c^2$

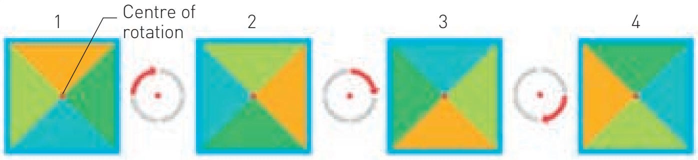
ROTATIONAL SYMMETRY

If a shape can be moved around a centre point and still fit its original outline exactly, it is said to have rotational symmetry. The order of rotational symmetry is the number of ways a shape can fit into its original outline when rotated.

EQUILATERAL TRIANGLE
An equilateral triangle has rotational symmetry of order 3 – when rotated, it fits its original outline in three different ways.



SQUARE
When rotated around its centre, a square fits its original outline in four different ways – its rotational symmetry is order 4.

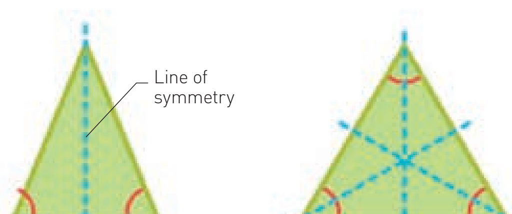


REFLECTIVE SYMMETRY

A reflection shows a shape in its mirror image, like a mountain reflection in a lake. When a flat shape can be divided in half so that each half is the exact mirror image of the other, it is said to have reflective symmetry. The line that divides the shape to perform the reflection is called a line of symmetry.

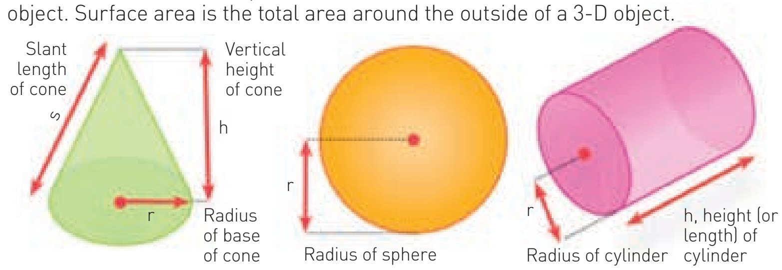
ISOSCELES TRIANGLE
This is symmetrical across a central line: the sides and angles on either side of the line are equal, and the line cuts the base in half at right angles.

EQUILATERAL TRIANGLE
An equilateral triangle has a line of symmetry through the middle of each side – not just the base.



FINDING VOLUME AND SURFACE AREA

Volume is the amount of space enclosed within a three-dimensional (3-D) object. Surface area is the total area around the outside of a 3-D object.



surface area = $\pi rs + \pi r^2$
volume = $\frac{1}{3}\pi r^2 h$

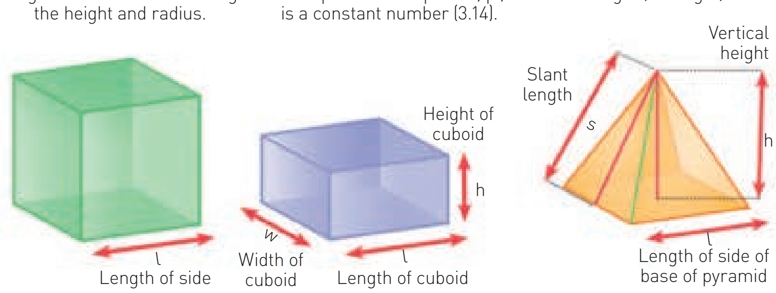
CONE
Find the surface area of a cone using the radius of its base, its height, and its slant length. Find the volume using the height and radius.

surface area = $4\pi r^2$
volume = $\frac{4}{3}\pi r^2$

SPHERE
You can find the surface area and volume of a sphere using only its radius, because the other part of the equation, pi, is a constant number (3.14).

surface area = $2\pi r(h+r)$
volume = $\pi r^2 h$

CYLINDER
The surface area and volume of a cylinder can be found from its radius and height (or length).



surface area = $6l^2$
volume = l^3

CUBE
The surface area and volume of a cube can be found by using only the length of its sides. No other information is needed.

surface area = $2(lh+lw+hw)$
volume = lwh

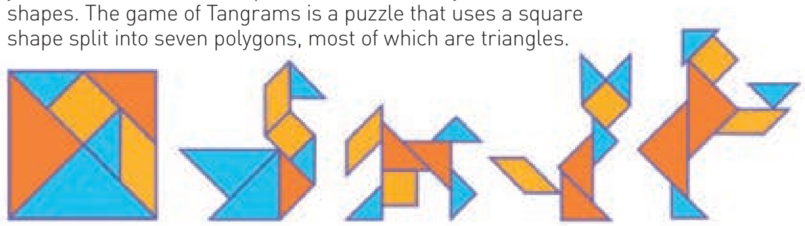
CUBOID
The surface area or volume of a cuboid can be found if you know its length, width, and height.

surface area = $2ls+l^2$
volume = $\frac{1}{3}l^2h$

SQUARE PYRAMID
Find the surface area of a square pyramid by using the lengths of its slant and the side of its base. Its volume can be found from its height and the side of its base.

TANGRAMS

Any shape that is made of straight sides can be split into triangles. If you cut up a piece of paper into triangles, for instance, you could reassemble the pieces in different ways to create new shapes. The game of Tangrams is a puzzle that uses a square shape split into seven polygons, most of which are triangles.



TANGRAM

POSSIBLE SHAPES USING TANGRAM PIECES